

Cubic Spectral Collapse for Endpoint-Spiked Hamiltonian Product Shears: Sharp Selector Thresholds in Arbitrary Mode Number

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Abstract

Let K be any field of characteristic zero, let $r \geq 4$, and let $g \geq 2r + 1$. We determine the degree growth of the polynomial symplectic automorphism obtained by composing the gradient shears generated by

$$V_{r,g}(q) = \prod_{i=1}^r q_i^2 + q_1^g, \quad W_{r,g}(p) = \prod_{i=1}^r p_i^2 + p_r^g.$$

Define

$$S(q, p) = (q, p + \nabla V_{r,g}(q)), \quad T(q, p) = (q + \nabla W_{r,g}(p), p), \quad F_{r,g} = T \circ S.$$

Put $h = g - 1$, let $M = 2\mathbf{1}\mathbf{1}^\top - I_r$, replace the first row of M by $he_1^\top$ to obtain A , replace the last row of M by $he_r^\top$ to obtain B , and set $C = BA$. An explicit sufficient invariant selector cone proves, for actual coordinate degrees, $v_{n+1} = Au_n$ and $u_{n+1} = Cu_n$. The proof includes both phase selectors, every cone wall, carried coordinates, and leading-form survival in a polynomial domain. The last q -coordinate has strict visibility for $n \geq 1$; the tied initial case at $n = 0$ is handled separately. Thus

$$\deg(F_{r,g}^n) = e_r^\top C^n \mathbf{1}, \quad \lambda_1(F_{r,g}) = \rho(C).$$

Middle-coordinate differences form an $(r - 3)$ -dimensional eigenspace on which C is the identity. Its invariant complement has dimension three, so, with $m = r - 2$,

$$\chi_C(t) = (t - 1)^{r-3} P_{m,h}(t), \quad P_{m,h}(1) = -4m(m + 1)(h + 1)^2 \neq 0,$$

where $P_{m,h}$ is an explicit cubic and a cubic annihilator of the exact degree sequence. The value $g = 2r$ is sharp only at the ordinary seed and the chosen strict selected face.

1 Introduction and bounded positioning

The concrete question in this article is whether one can read the degree of every iterate of a fixed two-shear word directly from its gradient supports. For the endpoint-spiked product potentials

$$V_{r,g}(q) = \prod_{i=1}^r q_i^2 + q_1^g, \quad W_{r,g}(p) = \prod_{i=1}^r p_i^2 + p_r^g, \quad (1.1)$$

the answer is not supplied by a formal exponent matrix alone. The pure power in the first V -derivative must beat a mixed product, and the pure power in the last W -derivative must win after the first phase has already changed the degree vector. At later iterates, each fresh gradient must also beat the coordinate carried from the preceding phase. Finally, even a unique degree score

would be inconclusive if its highest homogeneous form could vanish. These are separate obligations, resolved below in the order in which they occur in the map.

The payoff is a dimension-independent spectral statement. Although the degree vector has r entries, increasing r adds only middle-difference directions on which the complete-step matrix is the identity. All nontrivial growth remains in one explicitly identified three-dimensional subspace. The following theorem states the complete result before the bounded contextual discussion.

Theorem 1.1 (Main theorem). *Let K be a field of characteristic zero, and let $r, g \in \mathbb{Z}$ satisfy $r \geq 4$ and $g \geq 2r + 1$. Define*

$$S(q, p) = (q, p + \nabla V_{r,g}(q)), \quad T(q, p) = (q + \nabla W_{r,g}(p), p), \quad F_{r,g} = T \circ S,$$

where $V_{r,g}$ and $W_{r,g}$ are given by (1.1). Put $h = g - 1$, $m = r - 2$, and

$$M = 211^\top - I_r.$$

Let A be M with its first row replaced by $he_1^\top$, let B be M with its last row replaced by $he_r^\top$, and put $C = BA$. Then:

- (i) S , T , and $F_{r,g}$ are polynomial automorphisms preserving the standard symplectic form $\omega = \sum_{i=1}^r dq_i \wedge dp_i$.
- (ii) For $u > 0$, define $x_i = u_i/u_1$ only for $2 \leq i \leq r$ and $\sigma = \sum_{i=2}^r x_i$. The region

$$\mathcal{K}_{r,g} = \left\{ u > 0 : x_i \geq 1 \ (2 \leq i \leq r), \quad \sigma < \frac{h-1}{2} \right\}$$

is an explicit sufficient invariant selector cone containing $\mathbf{1}$. For the actual coordinate-degree vectors,

$$v_{n+1} = Au_n, \quad u_{n+1} = Cu_n \quad (n \geq 0).$$

- (iii) The last q -coordinate strictly dominates all other coordinate degrees for $n \geq 1$. At $n = 0$, all $2r$ coordinate degrees tie at one. For every $n \geq 0$,

$$\deg(F_{r,g}^n) = e_r^\top C^n \mathbf{1}, \quad \lambda_1(F_{r,g}) = \rho(C). \quad (1.2)$$

- (iv) For

$$U = \left\{ z \in K^r : z_1 = z_r = 0, \quad \sum_{i=2}^{r-1} z_i = 0 \right\}, \quad E = \left\{ (a, b, \dots, b, c)^\top : a, b, c \in K \right\},$$

one has $K^r = U \oplus E$, $\dim U = r - 3$, $A|_U = B|_U = -I_U$, and $C|_U = I_U$. In the convention $(a, b, c) \mapsto (a, b, \dots, b, c)$, with m equal middle coordinates,

$$Q_{m,h} = \begin{pmatrix} h + 4m + 4 & 2m(2m + 1) & 2(2m + 1) \\ 2h + 4m + 2 & 4m^2 + 1 & 4m \\ 2h & 2mh & h \end{pmatrix}$$

represents $C|_E$, and

$$\chi_C(t) = (t - 1)^{r-3} P_{m,h}(t), \quad (1.3)$$

where

$$\begin{aligned} P_{m,h}(t) = & t^3 - (2h + 4m^2 + 4m + 5)t^2 \\ & + (h^2 - 8hm(m+1) + 2h + 4)t \\ & - h^2(2m+1)^2. \end{aligned} \tag{1.4}$$

Moreover,

$$P_{m,h}(1) = -4m(m+1)(h+1)^2 \neq 0,$$

so the eigenvalue 1 has algebraic and geometric multiplicity exactly $r - 3$.

(v) If $d_n = e_r^T C^n \mathbf{1}$, then $P_{m,h}$ is a cubic annihilator:

$$d_{n+3} = T_0 d_{n+2} - S_0 d_{n+1} + D_0 d_n,$$

where

$$T_0 = 2h + 4m^2 + 4m + 5, \quad S_0 = h^2 - 8hm(m+1) + 2h + 4, \quad D_0 = h^2(2m+1)^2,$$

and $d_0 = 1$, $d_1 = h(2m+3)$. At $g = 2r$, the ordinary seed lies on the boundary of the selected open cone and the pure and mixed scores of the first selector tie; this is a seed/selected-face boundary only. The same conclusions hold for four arbitrary nonzero coefficients on the same fixed supports, as stated in Corollary 8.2.

The theorem has three falsifiable contributions.

- We derive both phase matrices from literal gradient supports and prove one explicit sufficient invariant selector cone, including both selectors, every cone wall, every carried coordinate, and leading-form survival.
- We prove the exact degree and Perron formulas, then decompose the r -dimensional degree system into an $(r - 3)$ -dimensional unit sector and the displayed three-dimensional equal-middle restriction.
- We compute the characteristic factor and cubic recurrence, prove exact unit multiplicity, and isolate the $g = 2r$ seed/selected-face boundary and the fixed-support coefficient corollary.

Why the matrix computation needs a proof chain

There are four points at which a plausible matrix calculation could fail to describe the polynomial iterates. First, each competitive derivative row contains two exponent vectors. Selecting the pure endpoint row at the ordinary seed does not show that the same row wins after the degree vector has moved. The open height face in $\mathcal{K}_{r,g}$ gives a uniform first selector, while the lower faces keep the ratios in a range on which that height comparison can be reused.

Second, the two endpoint comparisons occur at different times. The first one is evaluated on the q -degree vector u . The second one is evaluated on $v = Au$, because the W -gradient consumes the new p -coordinates produced by S . Replacing that second input by u would change its margin and would not analyze the displayed composition. The sign split in (3.13) and (3.14) is the algebraic record of this phase dependence.

Third, a gradient update is a sum of a carried coordinate and several fresh monomial sources. Even after one monomial is uniquely largest inside the gradient, the carried coordinate might match or exceed it. The inequalities $A\mathbf{1} > \mathbf{1}$, $C - I_r > 0$, and the nonnegative nonzero rows of A close

this separate comparison. They also explain why the induction is labelled by half-steps rather than written only for full iterates.

Fourth, degrees of summands do not certify survival of a polynomial leading form. A top-degree product must be nonzero, and a second source of the same degree could in principle cancel it. Strict uniqueness removes the second possibility, while the integral-domain identity $\text{LH}(fg) = \text{LH}(f)\text{LH}(g) \neq 0$ removes the first. This argument is robust under arbitrary nonzero scalar changes on the fixed supports, unlike an argument based only on positivity of coefficients.

Only after these four gaps are closed does the last coordinate become an observable for ordinary total degree. The invariant splitting then analyzes the spectrum of a matrix already known to govern actual degrees. This order prevents the cubic factor from being used retroactively as evidence for a support selection or for noncancellation.

Bounded context

Explicit degree calculations for affine-triangular automorphisms provide a broader comparison setting in arbitrary dimension, including realization questions for weak Perron numbers; dimension-four affine-triangular maps have also received a specialized algebraic-degree analysis (Blanc and van Santen, 2021; Shao and Sun, 2025). Those works motivate finite matrix descriptions but do not identify the displayed gradient family, select its two endpoint rows, or provide the cone used here. Every family-specific identity in Theorem 1.1 is proved from the two potentials.

Higher-dimensional polynomial automorphisms can exhibit degree-growth behavior not exhausted by the simplest planar alternatives (Déserti, 2016). This article addresses one narrow exponential mechanism rather than classifying those behaviors. Its main issue is the passage from formal support scores to actual coordinate degrees at every iterate.

Spectral interpretations of dynamical degrees provide general motivation for asking when a spectral radius is visible to a degree observable (Dang and Favre, 2021). Here visibility is proved by a last-coordinate comparison. The equality $\lambda_1(F_{r,g}) = \rho(C)$ follows from the exact degree formula, positivity of C , and a positive functional, not from contextual terminology.

Polynomial symplectic-map constructions and coupled Hénon systems form neighboring geometric settings (Rangarajan, 2002; Fujioka et al., 2023). Their construction and hyperbolicity questions differ from the support selection problem treated here. No hyperbolic, topological, or metric conclusion is imported into this algebraic-degree calculation.

Proof roadmap

The proof follows the composition word in twelve steps.

1. Section 2 expands the gradients and proves subtraction inverses and symplecticity.
2. Section 3 reads every support literally and derives M , A , B , and $C = BA$.
3. It then proves the first and second strict selectors in phase order.
4. Section 4 verifies the seed and every cone wall.
5. Section 5 treats the carried p -coordinates.
6. It treats the carried q -coordinates and proves leading-form survival in the polynomial domain.
7. Section 6 proves last-coordinate visibility for $n \geq 1$ and the tied $n = 0$ degree identity.

8. Positivity and visibility then give the Perron–Frobenius limit.
 9. Section 7 constructs $U \oplus E$ and computes Q .
 10. It derives the cubic, evaluates $P(1)$, proves unit multiplicity, and obtains the scalar recurrence.
 11. Section 8 audits $g = 2r$ and formal low-mode consistency.
 12. It proves the four-nonzero-coefficient fixed-support corollary and records the limitations.
- No later step certifies an earlier support choice.

2 The family, polynomial inverses, and symplectic geometry

Fix a field K of characteristic zero and integers $r \geq 4$, $g \geq 2r + 1$. Put $h = g - 1$ and $m = r - 2$, so $m \geq 2$ and $h \geq 2m + 4$. Use coordinates

$$q = (q_1, \dots, q_r), \quad p = (p_1, \dots, p_r)$$

on $\mathbb{A}_K^{2r}$. Ordinary total degree gives every initial coordinate degree one. For a polynomial map $G = (G_1, \dots, G_{2r})$, write $\deg(G) = \max_j \deg(G_j)$. Submultiplicativity under composition gives the algebraic limit

$$\lambda_1(G) = \lim_{n \rightarrow \infty} \deg(G^n)^{1/n}.$$

2.1 Fixed potentials and phase order

The two endpoint-spiked product potentials are

$$V_{r,g}(q) = \prod_{i=1}^r q_i^2 + q_1^g, \quad W_{r,g}(p) = \prod_{i=1}^r p_i^2 + p_r^g. \quad (2.1)$$

They define the block-triangular gradient shears

$$S(q, p) = (q, p + \nabla V_{r,g}(q)), \quad T(q, p) = (q + \nabla W_{r,g}(p), p), \quad (2.2)$$

and throughout the article the word is fixed as

$$F_{r,g} = T \circ S. \quad (2.3)$$

Thus S updates the p -block from the current q -block, and only then does T update the q -block from the newly produced p -block. Reversing the word would reverse the matrix product and would define a different problem.

The term *gradient* is used literally: the added vector is the gradient of the displayed polynomial. The maps in (2.2) are finite polynomial shears; no convergence, time parameter, or analytic flow is involved. All later degree comparisons take place in the coordinate ring of $\mathbb{A}_K^{2r}$.

Two degree vectors appear because a full iterate contains two triangular phases. If $f_1, \dots, f_r$ are nonzero coordinate polynomials with $\deg f_i = u_i$, a monomial $\prod_i f_i^{a_i}$ has candidate degree $\sum_i a_i u_i$. Its top homogeneous part is $\prod_i \text{LH}(f_i)^{a_i}$, so the candidate is the actual degree whenever that product is nonzero. Over a field, the polynomial ring is a domain, and the product is indeed nonzero. A sum of different support monomials still requires a strict comparison: two equal candidate degrees could interact. Sections 3 and 5 deliberately separate the score comparison from the leading-form argument.

The matrices below act on ordinary degree vectors, not on the coordinate polynomials themselves. Their entries are exponent data. The coefficient field enters in two other ways: derivative scalars must remain nonzero, and the direct-sum argument in Section 7 divides by the nonzero integer m . Characteristic zero is sufficient for both uses. Neither the inverse formulas nor the symplectic block calculation needs algebraic closedness.

The full map can be written explicitly as

$$F_{r,g}(q, p) = (q + \nabla W_{r,g}(p + \nabla V_{r,g}(q)), p + \nabla V_{r,g}(q)).$$

This formula makes the phase order visible: the second gradient is evaluated on the output of the first. It also shows why the p -degree vector after a full iterate is naturally the intermediate vector from the preceding half-step.

2.2 Gradients and derivative supports

Differentiating the two potentials gives

$$\frac{\partial V_{r,g}}{\partial q_1} = 2q_1 \prod_{j=2}^r q_j^2 + gq_1^{g-1}, \quad (2.4)$$

$$\frac{\partial V_{r,g}}{\partial q_i} = 2q_i \prod_{j \neq i} q_j^2 \quad (2 \leq i \leq r), \quad (2.5)$$

$$\frac{\partial W_{r,g}}{\partial p_i} = 2p_i \prod_{j \neq i} p_j^2 \quad (1 \leq i < r), \quad (2.6)$$

$$\frac{\partial W_{r,g}}{\partial p_r} = 2p_r \prod_{j=1}^{r-1} p_j^2 + gp_r^{g-1}. \quad (2.7)$$

Characteristic zero ensures that 2 and g are nonzero in K , so none of these supports disappears.

2.3 Polynomial inverses

The maps are triangular by blocks. Since S fixes q , and T fixes p , their inverses are

$$S^{-1}(q, p) = (q, p - \nabla V_{r,g}(q)), \quad T^{-1}(q, p) = (q - \nabla W_{r,g}(p), p). \quad (2.8)$$

Direct substitution verifies both identities. Thus S , T , and $F_{r,g} = T \circ S$ are polynomial automorphisms over K ; algebraic closedness is not used.

For completeness, composition in the reverse order of the inverse phases gives the full inverse

$$F_{r,g}^{-1}(q, p) = (q - \nabla W_{r,g}(p), p - \nabla V_{r,g}(q - \nabla W_{r,g}(p))). \quad (2.9)$$

Indeed, T^{-1} first recovers the old q -block while leaving p fixed, and S^{-1} then recovers the old p -block from that q -block. This formula is polynomial over K and uses no division.

2.4 Symplecticity

Let

$$\omega = \sum_{i=1}^r dq_i \wedge dp_i, \quad \Omega = \begin{pmatrix} 0 & I_r \\ -I_r & 0 \end{pmatrix}.$$

Writing $H_V = \nabla^2 V_{r,g}$ and $H_W = \nabla^2 W_{r,g}$, the Jacobians in the coordinate order (q, p) are

$$J_S = \begin{pmatrix} I_r & 0 \\ H_V & I_r \end{pmatrix}, \quad J_T = \begin{pmatrix} I_r & H_W \\ 0 & I_r \end{pmatrix}. \quad (2.10)$$

Both Hessians are symmetric. Block multiplication gives

$$J_S^\top \Omega J_S = \begin{pmatrix} H_V - H_V^\top & I_r \\ -I_r & 0 \end{pmatrix} = \Omega$$

and

$$J_T^\top \Omega J_T = \begin{pmatrix} 0 & I_r \\ -I_r & H_W^\top - H_W \end{pmatrix} = \Omega.$$

Equivalently, the added $dq_i \wedge dq_j$ and $dp_i \wedge dp_j$ terms cancel in symmetric pairs.

One can see the same cancellation directly. Under S ,

$$dp_i \longmapsto dp_i + \sum_j \frac{\partial^2 V_{r,g}}{\partial q_i \partial q_j} dq_j.$$

The extra contribution to $S^*\omega$ is a sum of $V_{ij} dq_i \wedge dq_j$. The terms for (i, j) and (j, i) cancel because $V_{ij} = V_{ji}$, and the diagonal wedge terms vanish. The pullback by T is the analogous calculation with p -variables and H_W . This coordinate check also shows that no analytic notion of a Hamiltonian flow is being used; the assertion is a polynomial identity over K .

Proposition 2.1. *The maps S , T , and $F_{r,g}$ are polynomial symplectic automorphisms of $\mathbb{A}_K^{2r}$.*

Proof. Polynomial invertibility follows from (2.8). The two block identities following (2.10) prove preservation of ω by each shear and hence by their composition. $\square$

Symplecticity fixes the geometric class of the maps but does not decide which monomial wins in either competitive derivative.

3 Gradient supports and the two strict selectors

Put

$$h = g - 1, \quad m = r - 2. \quad (3.1)$$

Then $m \geq 2$ and $h \geq 2m + 4$. Let $u = (u_1, \dots, u_r)^\top > 0$ be a degree vector. The score of a monomial with exponent row a is au .

3.1 Literal support rows

Every mixed derivative in (2.4)–(2.7) has exponent one in its own coordinate and exponent two in every other coordinate. Its score in row i is

$$u_i + 2 \sum_{j \neq i} u_j = 2 \sum_{j=1}^r u_j - u_i. \quad (3.2)$$

Consequently the mixed rows form

$$M = \mathbf{211}^\top - I_r. \quad (3.3)$$

Table 1: Gradient support-row ledger. Only the endpoint rows have a second competitor.

Potential	derivative rows	mixed support and score	additional pure support
$V_{r,g}$	$i = 1$	row $e_1^\top M$, score $u_1 + 2 \sum_{j=2}^r u_j$	$he_1^\top$, score hu_1
$V_{r,g}$	$2 \leq i \leq r$	row $e_i^\top M$, score $2 \sum_j u_j - u_i$	none
$W_{r,g}$	$1 \leq i < r$	row $e_i^\top M$, score $2 \sum_j u_j - u_i$	none
$W_{r,g}$	$i = r$	row $e_r^\top M$, score $2 \sum_{j < r} u_j + u_r$	$he_r^\top$, score hu_r

There are exactly two competitive rows. If their pure supports win, the phase matrices are

$$\begin{aligned}
A &= M + e_1(he_1^\top - e_1^\top M), \\
B &= M + e_r(he_r^\top - e_r^\top M), \\
C &= BA.
\end{aligned} \tag{3.4}$$

The order is forced by $F_{r,g} = T \circ S$: A acts first, and B acts on the resulting vector.

The replacement description can also be checked derivative by derivative. For $2 \leq i \leq r$, the i -th derivative of $V_{r,g}$ has only the mixed exponent row $e_i^\top M$; therefore the corresponding row of A cannot change with the input vector. Only the first derivative contains the additional exponent row $he_1^\top$. The roles are reversed for $W_{r,g}$: rows $1, \dots, r-1$ are forced mixed rows, and only row r can switch. Thus the matrix description has exactly two logical hypotheses. There is no unrecorded comparison in a middle row.

This literal ledger also explains why coefficients are absent from the matrix entries. A nonzero scalar changes a leading coefficient, not an exponent score. Coefficients can be ignored only after checking that they are nonzero and after proving that the selected leading form cannot cancel. Those checks are postponed to the phase induction, where the coordinate polynomials are available.

3.2 The normalized cone and first selector

For $u > 0$, define only

$$x_i = \frac{u_i}{u_1} \quad (2 \leq i \leq r), \quad \sigma = \sum_{i=2}^r x_i. \tag{3.5}$$

The index 1 is excluded from σ .

Definition 3.1. *The explicit sufficient invariant selector cone is*

$$\mathcal{K}_{r,g} = \left\{ u \in \mathbb{R}_{>0}^r : x_i \geq 1 \ (2 \leq i \leq r), \quad \sigma < \frac{h-1}{2} = \frac{g-2}{2} \right\}. \tag{3.6}$$

The lower inequalities alone imply

$$\sigma = x_2 + \dots + x_r \geq r - 1 = m + 1.$$

The parameter bound leaves genuine room above this lower corner, because

$$\frac{h-1}{2} - (m+1) = \frac{h-2m-3}{2} \geq \frac{1}{2}.$$

Thus the cone conditions are compatible already at the least allowed parameter. The same quantity $h - 2m - 3$ will appear in every reduced strict margin, making it possible to audit the threshold without replacing the open height inequality by a non-strict closure.

In the first V -row, pure minus mixed is

$$hu_1 - \left(u_1 + 2 \sum_{i=2}^r u_i \right) = u_1(h - 1 - 2\sigma) > 0. \quad (3.7)$$

Put $\delta = h - 1 - 2\sigma > 0$. The first pure support is uniquely selected, and all other V -rows have one support. For $v = Au$,

$$\frac{v_1}{u_1} = h, \quad \frac{v_i}{u_1} = 2 + 2\sigma - x_i \quad (2 \leq i \leq r). \quad (3.8)$$

3.3 The second selector after the first phase

The last pure support of the W -gradient is tested on $v = Au$, never directly on u . Its pure-minus-mixed margin is

$$\Delta_T = (h - 1)v_r - 2 \sum_{j < r} v_j. \quad (3.9)$$

There are m middle indices before r , and

$$\frac{1}{u_1} \sum_{j < r} v_j = h + \sum_{j=2}^{r-1} (2 + 2\sigma - x_j) = h + 2m + (2m - 1)\sigma + x_r.$$

Also $v_r/u_1 = 2 + 2\sigma - x_r$. Substitution in (3.9) gives

$$\frac{\Delta_T}{u_1} = (2h - 4m)\sigma - (h + 1)x_r - 4m - 2. \quad (3.10)$$

There are m middle variables, each at least one, so

$$x_r \leq \sigma - m. \quad (3.11)$$

Indeed,

$$\sigma - x_r = \sum_{i=2}^{r-1} x_i \geq m.$$

This inequality retains the last ratio as the distinguished endpoint variable; it does not symmetrize x_r with the m middle ratios. Therefore

$$\frac{\Delta_T}{u_1} \geq (h - 4m - 1)\sigma + m(h - 3) - 2. \quad (3.12)$$

If $h - 4m - 1 < 0$, multiplying $\sigma < (h - 1)/2$ by that negative coefficient reverses the inequality:

$$\begin{aligned} \frac{\Delta_T}{u_1} &> \frac{(h - 4m - 1)(h - 1)}{2} + m(h - 3) - 2 \\ &= \frac{(h + 1)(h - 2m - 3)}{2} > 0. \end{aligned} \quad (3.13)$$

If $h - 4m - 1 \geq 0$, the lower bound $\sigma \geq m + 1$ gives

$$\begin{aligned} \frac{\Delta_T}{u_1} &\geq (h - 4m - 1)(m + 1) + m(h - 3) - 2 \\ &= (2m + 1)(h - 2m - 3) > 0. \end{aligned} \quad (3.14)$$

Both cases are strict because $h - 2m - 3 \geq 1$.

At $h = 2m + 4$, the common decisive factor is exactly one. The remaining factors in the two bounds are $h + 1 > 0$ and $2m + 1 > 0$, respectively. Hence the endpoint selector is already strict at the least parameter; no unwritten strengthening of $g \geq 2r + 1$ is used. The alternatives $h - 4m - 1 < 0$ and $h - 4m - 1 \geq 0$ are exhaustive, including the zero coefficient in the second case.

Proposition 3.2 (Strict phase selectors). *For $u \in \mathcal{K}_{r,g}$, the first gradient phase uniquely selects the rows of A . For $v = Au$, the second phase uniquely selects the rows of B . These are the only two competitions.*

Proof. The first margin is (3.7). The second is (3.13) or (3.14). The literal ledger shows that every other derivative row has one support. $\square$

The two sign cases are not a partition of the parameter range into different theorems. They identify which boundary of the same cone minimizes a linear margin. When the coefficient of σ is negative, the infimum is approached at the open height face, and the strict input inequality is what produces a strict lower bound. When it is nonnegative, the lower faces $x_i \geq 1$ supply the minimum. Retaining both cases avoids replacing a valid inequality by one with the wrong direction near the height boundary.

4 Seed containment and strict cone invariance

The selectors apply at a single input vector. To iterate them, one must show that a complete phase pair returns the vector to the same cone. This requires the full product $C = BA$.

4.1 The complete-step rows

We multiply B and A row by row. The first row of B has entries $1, 2, \dots, 2$, while the first column of A is $(h, 2, \dots, 2)^\top$. Hence

$$C_{11} = h + 4(r - 1) = h + 4m + 4.$$

For $j \geq 2$, the sum of column j of A over rows $2, \dots, r$ is $1 + 2(r - 2) = 2m + 1$, so $C_{1j} = 4m + 2$.

For a middle row $2 \leq i < r$, the first entry of the B -row is two, and the sum of its remaining entries is $2m + 1$. Thus $C_{i1} = 2h + 4m + 2$. For $j \geq 2$,

$$C_{ij} = \sum_{k=2}^r (2 - \mathbf{1}_{\{i=k\}})(2 - \mathbf{1}_{\{k=j\}}) = 4m + \mathbf{1}_{\{i=j\}}.$$

The last row of B is $he_r^\top$, while the last row of A is the last row of M , giving $C_{rj} = 2h$ for $j < r$ and $C_{rr} = h$. The complete entry ledger is therefore

$$\begin{aligned} C_{11} &= h + 4m + 4, & C_{1j} &= 4m + 2 & (2 \leq j \leq r), \\ C_{i1} &= 2h + 4m + 2 & (2 \leq i < r), & C_{ij} &= 4m + \mathbf{1}_{\{i=j\}} & (2 \leq i < r, 2 \leq j \leq r), \\ C_{rj} &= 2h & (1 \leq j < r), & C_{rr} &= h. \end{aligned} \tag{4.1}$$

For a middle row, normalization can be seen directly:

$$\frac{(Cu)_i}{u_1} = 2h + 4m + 2 + \sum_{j=2}^r (4m + \mathbf{1}_{\{i=j\}})x_j = 2h + 4m + 2 + 4m\sigma + x_i.$$

For the last row,

$$\frac{(Cu)_r}{u_1} = 2h \left(1 + \sum_{j=2}^{r-1} x_j \right) + hx_r = h(2 + 2\sigma - x_r).$$

After division by u_1 , the three row types are

$$\frac{(Cu)_1}{u_1} = h + 4m + 4 + (4m + 2)\sigma, \quad (4.2)$$

$$\frac{(Cu)_i}{u_1} = 2h + 4m + 2 + 4m\sigma + x_i \quad (2 \leq i < r), \quad (4.3)$$

$$\frac{(Cu)_r}{u_1} = h(2 + 2\sigma - x_r). \quad (4.4)$$

4.2 The normalized complete-step map

The row formulas also give the complete normalized transformation explicitly. Set

$$D = h + 4m + 4 + (4m + 2)\sigma = \frac{(Cu)_1}{u_1}. \quad (4.5)$$

Because $D > 0$, the ratios of Cu relative to its first coordinate are

$$x'_i = \frac{2h + 4m + 2 + 4m\sigma + x_i}{D} \quad (2 \leq i < r), \quad (4.6)$$

$$x'_r = \frac{h(2 + 2\sigma - x_r)}{D}. \quad (4.7)$$

Thus every output inequality can be checked without an implicit projectivization convention. The lower wall $x'_i \geq 1$ is equivalent to $(Cu)_i - (Cu)_1 \geq 0$, since the common denominator is positive. The output height inequality $\sum_{i=2}^r x'_i < (h - 1)/2$ is equivalent to

$$(h - 1)(Cu)_1 - 2 \sum_{i=2}^r (Cu)_i > 0.$$

These are exactly the quantities treated below.

For reference, summing (4.6) over the m middle indices and then adding (4.7) gives

$$\sum_{i=2}^r x'_i = \frac{m(2h + 4m + 2 + 4m\sigma) + (\sigma - x_r) + h(2 + 2\sigma - x_r)}{D}. \quad (4.8)$$

No estimate has entered this identity. It makes clear that the normalized system has precisely $m + 1$ lower walls and one height wall: the denominator introduces no new sign condition, and the radial coordinate has disappeared. The proof below works with the cross-multiplied forms because their factorizations expose the sharp parameter $h - 2m - 3$.

4.3 Seed containment

At the ordinary seed $u_0 = \mathbf{1}$, all ratios equal one and

$$\sigma(\mathbf{1}) = r - 1 = m + 1. \quad (4.9)$$

Since $h \geq 2m + 4$,

$$m + 1 < \frac{2m + 3}{2} \leq \frac{h - 1}{2}. \quad (4.10)$$

Thus $\mathbf{1}$ lies strictly in $\mathcal{K}_{r,g}$.

4.4 Every lower wall

Let $U' = Cu$. For $2 \leq i < r$, subtracting (4.2) from (4.3) gives

$$\frac{U'_i - U'_1}{u_1} = (h - 1 - 2\sigma) + (x_i - 1) > 0. \quad (4.11)$$

The first summand is the strict selector margin, and the second is nonnegative. Hence every middle input lower face maps strictly above its output lower face.

For the last coordinate,

$$L_1 := \frac{U'_r - U'_1}{u_1} = h - 4m - 4 + (2h - 4m - 2)\sigma - hx_r. \quad (4.12)$$

Using $x_r \leq \sigma - m$ gives

$$L_1 \geq (m + 1)(h - 4) + (h - 4m - 2)\sigma. \quad (4.13)$$

If $h - 4m - 2 < 0$, the open height inequality reverses after multiplication:

$$\begin{aligned} L_1 &> (m + 1)(h - 4) + \frac{(h - 4m - 2)(h - 1)}{2} \\ &= \frac{(h + 2)(h - 2m - 3)}{2} > 0. \end{aligned} \quad (4.14)$$

If $h - 4m - 2 \geq 0$, the lower bound $\sigma \geq m + 1$ gives

$$\begin{aligned} L_1 &\geq (m + 1)(h - 4) + (h - 4m - 2)(m + 1) \\ &= 2(m + 1)(h - 2m - 3) > 0. \end{aligned} \quad (4.15)$$

The last lower face therefore also maps strictly above one.

4.5 The open height wall

The output height inequality is equivalent, because $U'_1 > 0$, to positivity of

$$H_2 = (h - 1)U'_1 - 2 \sum_{i=2}^r U'_i. \quad (4.16)$$

Substitution of the three normalized rows gives

$$\frac{H_2}{u_1} = h^2 - h - 8m^2 - 8m - 4 + c\sigma + 2(h + 1)x_r, \quad (4.17)$$

where

$$c = (4m - 2)h - 8m^2 - 4m - 4. \quad (4.18)$$

For clarity, the collection in (4.17) uses

$$\sum_{i=2}^{r-1} x_i = \sigma - x_r$$

and exactly m middle rows. Before final collection, the normalized height form is

$$\frac{H_2}{u_1} = (h - 1)(h + 4m + 4 + (4m + 2)\sigma)$$

$$\begin{aligned}
& -2m(2h + 4m + 2 + 4m\sigma) - 2(\sigma - x_r) \\
& -2h(2 + 2\sigma - x_r).
\end{aligned}$$

The coefficient of x_r becomes $2 + 2h = 2(h + 1)$; the remaining ratio terms combine into $c\sigma$. This intermediate line accounts for all $m + 1$ output ratios and prevents the last coordinate from being counted twice.

At $h = 2m + 4$, one has $c = 8m - 12 > 0$, and c increases with h . Both variable coefficients in (4.17) are positive. Using $\sigma \geq m + 1$ and $x_r \geq 1$, exact collection yields

$$\begin{aligned}
\frac{H_2}{u_1} & \geq h^2 - h - 8m^2 - 8m - 4 + c(m + 1) + 2(h + 1) \\
& = (h - 2m - 3)(h + 4m^2 + 4m + 2) > 0.
\end{aligned} \tag{4.19}$$

This is precisely the strict output height condition. At the least parameter $h = 2m + 4$, every decisive factor $h - 2m - 3$ equals one; there is no hidden threshold equality.

Table 2: Selector and cone proof ledger. Alternatives correspond to the sign of the relevant linear coefficient.

Obligation	exact expression	inequality used	strict lower bound
first selector	$u_1(h - 1 - 2\sigma)$	$\sigma < (h - 1)/2$	> 0
second selector	(3.10)	$x_r \leq \sigma - m$, sign split	$\frac{(h+1)(h-2m-3)}{2}$ or $(2m + 1)(h - 2m - 3)$
seed	$\sigma(\mathbf{1}) = m + 1$	$h \geq 2m + 4$	slack at least $1/2$
middle lower wall	$(h - 1 - 2\sigma) + (x_i - 1)$	$\sigma < (h - 1)/2$, $x_i \geq 1$	> 0
last lower wall	(4.12)	$x_r \leq \sigma - m$, sign split	$\frac{(h+2)(h-2m-3)}{2}$ or $2(m + 1)(h - 2m - 3)$
height wall	(4.16)	$\sigma \geq m + 1$, $x_r \geq 1$, $c > 0$	$(h - 2m - 3)(h + 4m^2 + 4m + 2)$
least parameter	$h = 2m + 4$	$m \geq 2$	$h - 2m - 3 = 1$

Theorem 4.1 (Strict cone invariance). *The seed lies in $\mathcal{K}_{r,g}$, both selectors are strict there, and*

$$CK_{r,g} \subset \mathcal{K}_{r,g}. \tag{4.20}$$

Every output lower inequality and the output height inequality is strict.

Proof. Seed containment is (4.10). The selectors are Proposition 3.2. The middle, last, and height output conditions are respectively (4.11), (4.14)–(4.15), and (4.19). $\square$

The theorem proves sufficiency and invariance. It does not say that the cone is maximal, necessary, unique, exact, or a classification.

The wall audit is complete because the cone is homogeneous. Division by the positive first coordinate removes the radial direction, leaving the closed lower inequalities $x_i \geq 1$ and one open affine height inequality. Equation (4.11) treats each of the m middle lower faces, the two-case calculation treats the last lower face, and (4.19) treats the only upper face. Positivity of the first output coordinate is automatic from (4.2). No additional normalized wall remains.

Strict inward movement is stronger than mere invariance and is used twice later. It keeps the selectors away from a tie at every finite iterate, and the last-lower-wall inequality becomes the comparison between the last and first q -degrees in Section 6. The same symbolic margin therefore serves two logically distinct conclusions only after each has been stated with its proper input.

5 Carried coordinates and leading-form survival

The cone selects a support within each fresh gradient. Actual coordinate updates also contain an old coordinate, so the fresh degree must beat that carried term. Once it does, the uniquely highest source still needs a nonzero leading form.

5.1 Phase-labelled vectors and carried terms

Write

$$F_{r,g}^n(q, p) = (q^{(n)}, p^{(n)})$$

and let

$$u_n = (\deg q_1^{(n)}, \dots, \deg q_r^{(n)})^\top.$$

After applying S to the n -th full iterate, let v_{n+1} be the degree vector of

$$p^{(n)} + \nabla V_{r,g}(q^{(n)}).$$

Since T fixes that p -block, v_{n+1} is also the p -degree vector after the next full iterate. Initially $u_0 = \mathbf{1}$, and the initial p -degree vector is $\mathbf{1}$.

There are two maxima in each half-step. For the i -th coordinate of the first phase, the degree of

$$p_i^{(n)} + \frac{\partial V_{r,g}}{\partial q_i}(q^{(n)})$$

is the larger of the carried degree and the winning gradient degree, once leading-form survival is known. Proposition 3.2 determines the winner inside the gradient, but says nothing about $p_i^{(n)}$. Likewise, the i -th coordinate of the second phase compares $q_i^{(n)}$ with the winning degree in $\partial W_{r,g}/\partial p_i$, evaluated at the new p -block. This is why one needs both inequalities in Lemma 5.1, row by row.

The entry ledger shows that $C - I_r$ is entrywise positive. In the first row, subtraction leaves diagonal entry $h + 4m + 3$ and other entries $4m + 2$. In a middle row it leaves first entry $2h + 4m + 2$ and every entry in columns $2, \dots, r$ equal to $4m$. In the last row it leaves entries $2h$ before the final position and $h - 1$ at that position. Thus

$$Cu > u \quad (u > 0). \quad (5.1)$$

At the seed,

$$A\mathbf{1} = (h, 2r - 1, \dots, 2r - 1)^\top = (h, 2m + 3, \dots, 2m + 3)^\top > \mathbf{1}. \quad (5.2)$$

Every row of A is nonnegative and nonzero.

Lemma 5.1 (Carried-coordinate gaps). *If $u_n = Cu_{n-1}$ and $v_n = Au_{n-1}$ for $n \geq 1$, then*

$$Au_n > Au_{n-1} = v_n \quad (5.3)$$

and

$$BAu_n - u_n = (C - I_r)u_n > 0. \quad (5.4)$$

Proof. Equation (5.1) gives $u_n > u_{n-1}$. Each nonnegative nonzero row of A preserves strict componentwise growth, proving (5.3). Positivity of $C - I_r$ proves (5.4). $\square$

Equation (5.2) handles the first carried p -block, and (5.4) with $u_0 = \mathbf{1}$ handles the first carried q -block. Lemma 5.1 supplies both gaps at later phases.

5.2 Leading forms in a polynomial domain

Let

$$R = K[q_1, \dots, q_r, p_1, \dots, p_r].$$

For nonzero $f \in R$, denote its highest-total-degree homogeneous form by $\text{LH}(f)$.

Lemma 5.2 (Leading-form survival). *For nonzero $f, g \in R$,*

$$\text{LH}(fg) = \text{LH}(f) \text{LH}(g) \neq 0. \quad (5.5)$$

If $\deg f > \deg g$, then

$$\text{LH}(f + g) = \text{LH}(f). \quad (5.6)$$

Proof. The component of degree $\deg f + \deg g$ in fg is $\text{LH}(f) \text{LH}(g)$. The polynomial ring is an integral domain, so this product is nonzero. If $\deg f > \deg g$, no term of g has the top degree of f , proving the sum identity. $\square$

In an S -phase, Proposition 3.2 gives one uniquely highest derivative support. After substitution, its top form is a nonzero scalar times a product of powers of prior nonzero leading forms. The scalar is 2 or g , nonzero in characteristic zero, and (5.5) makes the product nonzero. The carried p -coordinate has strictly smaller degree by (5.2) or (5.3), so it cannot cancel that form.

The T -phase is identical in logic but uses the second selector on $v = Au$. Its selected product survives by the domain property, and the old q -coordinate has smaller degree by (5.4). No positive-coefficient assumption is used.

To see the induction at the level of an individual support, consider a mixed row i . Its substituted monomial has top form

$$2 \text{LH}(f_i) \prod_{j \neq i} \text{LH}(f_j)^2,$$

where the f_j are the current coordinates in the relevant block. Each $\text{LH}(f_j)$ is nonzero by induction, so the product is nonzero. In an endpoint pure row, the corresponding form is

$$g \text{LH}(f_1)^h \quad \text{or} \quad g \text{LH}(f_r)^h.$$

The selector makes exactly one of the mixed and pure forms higher in degree; they are never added at equal top degree on the cone. The carry comparison then makes the entire winning gradient form higher than the old coordinate. Thus every use of (5.6) has a strict premise.

This separation also handles possible dependence among the coordinate polynomials. The factors need not involve disjoint variables, nor must their coefficients have a common sign. The integral-domain property alone ensures that a product of their nonzero highest forms is nonzero. No monomial-order refinement or coefficient ordering is required.

Theorem 5.3 (Exact phase recurrence). *For every $n \geq 0$, the actual coordinate-degree vectors satisfy*

$$v_{n+1} = Au_n, \quad u_{n+1} = Cu_n. \quad (5.7)$$

Consequently,

$$u_n = C^n \mathbf{1} \quad (n \geq 0), \quad v_n = AC^{n-1} \mathbf{1} \quad (n \geq 1), \quad (5.8)$$

and $u_n \in \mathcal{K}_{r,g}$ for every n .

Proof. We use a simultaneous half-step induction. Its assertions at full time n are:

- (a) every $q_i^{(n)}$ and $p_i^{(n)}$ has a nonzero highest homogeneous form;
- (b) the q -degree vector is $u_n = C^n \mathbf{1}$ and belongs to $\mathcal{K}_{r,g}$;
- (c) for $n \geq 1$, the p -degree vector is $v_n = AC^{n-1} \mathbf{1}$.

At time zero, all coordinate polynomials are variables, so their highest forms are nonzero, $u_0 = \mathbf{1}$, and seed containment gives $u_0 \in \mathcal{K}_{r,g}$.

Apply S first. Proposition 3.2 chooses the row of A in every derivative. In a mixed row, the selected substituted highest form is

$$2 \text{LH}(q_i^{(0)}) \prod_{j \neq i} \text{LH}(q_j^{(0)})^2;$$

in the first row it is $g \text{LH}(q_1^{(0)})^h$. Both are nonzero. Their degrees form Au_0 , while (5.2) says that they strictly exceed the coordinatewise degrees of the carried initial p -block. Lemma 5.2 therefore shows that the new p -coordinates have nonzero highest forms and degree vector

$$v_1 = Au_0.$$

The second selector is now evaluated on this new block, as required by the composition order. It chooses the rows of B , and the same product argument makes every selected W -gradient form nonzero. The fresh degree vector is $Bv_1 = Cu_0$. Since $(C - I_r)u_0 > 0$, each fresh degree strictly exceeds the degree of the corresponding carried $q_i^{(0)}$. Another application of the sum part of Lemma 5.2 gives

$$u_1 = Cu_0$$

and nonzero highest forms in the new q -block. The T -phase leaves the new p -block fixed. Finally, strict cone invariance places u_1 back in $\mathcal{K}_{r,g}$. This proves all three assertions at time one and also records both half-steps of the base case.

Suppose now that the assertions hold at some $n \geq 1$. The first selector applies because $u_n \in \mathcal{K}_{r,g}$. It gives selected gradient degrees Au_n , with nonzero selected highest forms by the domain product identity. The carried p -degree vector is

$$v_n = Au_{n-1}.$$

Moreover $u_n = Cu_{n-1} > u_{n-1}$ by (5.1), and every row of A is nonnegative and nonzero. Hence

$$Au_n > Au_{n-1} = v_n.$$

The carried term is strictly lower in every coordinate, so after the S -phase the new p -degree vector is exactly

$$v_{n+1} = Au_n,$$

and all its highest forms remain nonzero.

For the following T -phase, the second selector applies to $v_{n+1} = Au_n$ and chooses B . Its selected gradient degrees are

$$Bv_{n+1} = BAu_n = Cu_n.$$

They strictly exceed the carried q -degrees u_n , because $(C - I_r)u_n > 0$. Their selected highest forms are again nonzero products of the just-established p -highest forms. Thus no selected term can vanish and no carried coordinate can cancel it. The new q -degree vector is

$$u_{n+1} = Cu_n,$$

all q -highest forms are nonzero, and cone invariance gives $u_{n+1} \in \mathcal{K}_{r,g}$. This closes the simultaneous induction. Iterating the two vector identities yields (5.8). $\square$

The induction proves more than the scalar degree of the full map. It records the degree of every coordinate after each half-step, preserves a strict selector certificate for the next input, and supplies nonzero leading forms for the next product substitution. These three outputs are all needed to repeat the argument. Keeping them together rules out a circular proof in which the matrix recurrence is assumed in order to establish the cone that was supposed to justify the recurrence.

The recurrence now describes actual coordinate degrees. Ordinary map degree still requires a coordinate that sees the maximum.

6 Last-coordinate visibility, exact degrees, and Perron growth

At a full iterate, the q -degree vector is Cu and the accompanying p -degree vector is Au . We compare those blocks coordinatewise and then compare the last q -row with every other q -row.

6.1 Full-step versus half-step degrees

Direct subtraction shows that $C - A$ is entrywise positive. Its first row is

$$(4m + 4, 4m + 2, \dots, 4m + 2).$$

For a middle row $2 \leq i < r$, its first entry is $2h + 4m$, and its entry in column $j \geq 2$ is

$$4m - 2 + 2\mathbf{1}_{\{i=j\}}.$$

Its last row has entries $2h - 2$ in columns $1, \dots, r - 1$ and $h - 1$ in the final column. All are positive for $m \geq 2$ and $h \geq 2m + 4$. Hence

$$Cu > Au \quad (u > 0). \tag{6.1}$$

Every full-step q_i -degree therefore exceeds the intermediate p_i -degree with the same index.

This indexwise comparison suffices for the whole p -block once the largest q -row is known. If q_r exceeds every q_i , and each q_i exceeds the corresponding p_i , then q_r exceeds every p_i as well. Thus the remaining work is entirely a comparison among the rows of C ; no separate last-row comparison with every row of A is needed.

6.2 The last row versus all other rows

For $u \in \mathcal{K}_{r,g}$, the inequality $(Cu)_r > (Cu)_1$ is exactly the last-lower-wall calculation (4.14)–(4.15). Fix a middle index $2 \leq i < r$ and put

$$s = \sum_{j=2}^{r-1} x_j.$$

Then $s \geq m$, $\sigma = s + x_r$, and the open height condition gives

$$x_r < \frac{h - 1}{2} - s. \tag{6.2}$$

Subtracting the middle normalized row from the last one gives

$$L_i := \frac{(Cu)_r - (Cu)_i}{u_1}$$

$$\begin{aligned}
&= (2h - 4m)\sigma - hx_r - x_i - 4m - 2 \\
&= (2h - 4m)s + (h - 4m)x_r - x_i - 4m - 2.
\end{aligned} \tag{6.3}$$

If $h - 4m < 0$, multiplication of (6.2) by this negative number reverses the inequality:

$$L_i > hs - x_i + \frac{(h - 4m)(h - 1)}{2} - 4m - 2. \tag{6.4}$$

Moreover,

$$\begin{aligned}
hs - x_i &= (h - 1)x_i + h \sum_{\substack{2 \leq j < r \\ j \neq i}} x_j \\
&\geq (h - 1) + h(m - 1) = hm - 1.
\end{aligned} \tag{6.5}$$

Therefore

$$L_i > \frac{(h + 2)(h - 2m - 3)}{2} > 0. \tag{6.6}$$

If $h - 4m \geq 0$, every coefficient in (6.3) is minimized on the lower faces $x_2 = \dots = x_r = 1$. The coefficient of x_i is $2h - 4m - 1 > 0$, the other middle coefficients are $2h - 4m > 0$, and the x_r -coefficient is nonnegative. Substitution yields

$$L_i \geq (2m + 1)(h - 2m - 3) > 0. \tag{6.7}$$

Thus the last component of Cu strictly exceeds every other component.

Proposition 6.1 (Strict visibility for $n \geq 1$). *For every $n \geq 1$, the last q -coordinate of $F_{r,g}^n$ has degree strictly larger than each of the other $2r - 1$ coordinate degrees.*

Proof. The q -degree vector is $u_n = C(C^{n-1}\mathbf{1})$, and the p -degree vector is $v_n = A(C^{n-1}\mathbf{1})$. Cone invariance puts $C^{n-1}\mathbf{1}$ in $\mathcal{K}_{r,g}$. The last-wall and (6.6)–(6.7) comparisons make the last component of u_n strictly largest in its block. Equation (6.1) gives $u_n > v_n$, completing the comparison with the p -block. $\square$

6.3 Exact ordinary degree and the initial tie

For $n \geq 1$, Proposition 6.1 gives

$$\deg(F_{r,g}^n) = e_r^\top C^n \mathbf{1}. \tag{6.8}$$

The restriction $n \geq 1$ applies to strict visibility, not to the formula. At $n = 0$, every coordinate degree is one, and

$$e_r^\top C^0 \mathbf{1} = e_r^\top \mathbf{1} = 1.$$

Theorem 6.2 (Exact degrees). *For every $n \geq 0$,*

$$\deg(F_{r,g}^n) = e_r^\top C^n \mathbf{1}. \tag{6.9}$$

Strict visibility holds for $n \geq 1$, whereas $n = 0$ is the tied initial case.

6.4 Perron visibility

The degree matrix has positive integer entries, so Perron–Frobenius is applied to C as a real matrix; no ordering or embedding of the coefficient field K is involved. Positivity makes C primitive. Hence $\rho(C) > 0$ is a simple eigenvalue, it has positive right and left eigenvectors, and every other eigenvalue has modulus strictly below $\rho(C)$.

Choose positive Perron eigenvectors w and ℓ , normalized by $\ell^\top w = 1$. The rank-one spectral projection onto the Perron line is $w\ell^\top$. Put

$$\kappa = (e_r^\top w)(\ell^\top \mathbf{1}).$$

This scalar is positive because the last-coordinate functional, the seed, and both Perron vectors have positive entries. Therefore there is a real number $0 < \theta < \rho(C)$ such that

$$e_r^\top C^n \mathbf{1} = \kappa \rho(C)^n + O(\theta^n). \quad (6.10)$$

If a subdominant Jordan block is present, θ may be chosen strictly between its spectral modulus and $\rho(C)$, absorbing the polynomial factor in the usual way.

Thus the observable used in the exact degree formula cannot annihilate the Perron component. Dividing (6.10) by $\rho(C)^n$ gives a positive limiting coefficient, and taking n -th roots yields

$$\lim_{n \rightarrow \infty} \left(e_r^\top C^n \mathbf{1} \right)^{1/n} = \rho(C).$$

Combining this limit with Theorem 6.2 proves:

Corollary 6.3 (First dynamical degree).

$$\lambda_1(F_{r,g}) = \rho(C). \quad (6.11)$$

This is an algebraic-degree statement. It is not an equality with topological, metric, arithmetic, or measure-theoretic entropy.

7 Unit modes, the equal-middle quotient, and cubic recurrence

The middle rows of C are permutation-symmetric. Difference vectors among the middle coordinates separate from the three directions recording the first endpoint, a common middle value, and the last endpoint.

7.1 The unit space and direct complement

Define

$$U = \left\{ z \in K^r : z_1 = z_r = 0, \quad \sum_{i=2}^{r-1} z_i = 0 \right\}. \quad (7.1)$$

There are m middle coordinates and one relation, so

$$\dim_K U = m - 1 = r - 3. \quad (7.2)$$

Every $z \in U$ has total coordinate sum zero; therefore $Mz = -z$. The replaced first row of A evaluates to $hz_1 = 0$, which equals $-z_1$, and the replaced last row of B evaluates to $hz_r = 0$. Thus

$$Az = -z, \quad Bz = -z, \quad Cz = z \quad (z \in U). \quad (7.3)$$

Concretely, U is spanned by differences of middle coordinate vectors, for example

$$e_2 - e_3, e_3 - e_4, \dots, e_{r-2} - e_{r-1}.$$

There are $r - 3$ independent differences. Each phase reverses them, while a full step restores them. This explains why adding a middle mode adds a unit direction rather than another nontrivial spectral coordinate.

Let

$$E = \left\{ (a, b, \dots, b, c)^\top : a, b, c \in K \right\}, \quad (7.4)$$

with exactly m middle copies of b . A vector in $U \cap E$ has $a = c = 0$ and $mb = 0$. Since $m \neq 0$ is invertible in the characteristic-zero field, $b = 0$. Dimension counting gives the invariant direct sum

$$K^r = U \oplus E. \quad (7.5)$$

The matrices preserve E because their middle rows are symmetric under permutations of the middle coordinates.

The decomposition also has an explicit projection. Given $z = (z_1, \dots, z_r)^\top \in K^r$, set

$$\bar{z} = \frac{1}{m} \sum_{i=2}^{r-1} z_i$$

and define

$$z_E = (z_1, \bar{z}, \dots, \bar{z}, z_r)^\top, \quad z_U = z - z_E.$$

Then $z_E \in E$, while z_U has zero endpoint coordinates and its middle coordinates sum to zero, so $z_U \in U$. This gives existence of the splitting without relying only on dimensions; the earlier intersection calculation gives uniqueness. The factor $1/m$ is the precise reason for retaining the characteristic-zero assumption in this linear-algebra step.

Middle-coordinate permutations preserve both subspaces. More concretely, the output of any middle row of A or B depends on its distinguished middle input and on the sum of all inputs. Equal middle inputs therefore produce equal middle outputs, proving $A(E) \subseteq E$ and $B(E) \subseteq E$ directly. Together with (7.3), this shows that the displayed direct sum is simultaneously invariant under both half-step matrices, not merely under their product.

7.2 The fixed equal-middle convention

Fix the coordinate convention

$$(a, b, c) \mapsto (a, \underbrace{b, \dots, b}_{m \text{ entries}}, c). \quad (7.6)$$

Applying A gives first coordinate ha , any middle coordinate

$$2(a + mb + c) - b = 2a + (2m - 1)b + 2c,$$

and last coordinate

$$2(a + mb + c) - c = 2a + 2mb + c.$$

Hence

$$A_E = \begin{pmatrix} h & 0 & 0 \\ 2 & 2m - 1 & 2 \\ 2 & 2m & 1 \end{pmatrix}. \quad (7.7)$$

For B , the first and middle rows retain their mixed form and the last row is pure:

$$B_E = \begin{pmatrix} 1 & 2m & 2 \\ 2 & 2m-1 & 2 \\ 0 & 0 & h \end{pmatrix}. \quad (7.8)$$

Multiplication in phase order yields

$$Q_{m,h} = B_E A_E = \begin{pmatrix} h+4m+4 & 2m(2m+1) & 2(2m+1) \\ 2h+4m+2 & 4m^2+1 & 4m \\ 2h & 2mh & h \end{pmatrix}. \quad (7.9)$$

For example,

$$(1, 2m, 2)A_E = (h+4m+4, 2m(2m+1), 2(2m+1)),$$

which gives its first row. The matrix $Q_{m,h}$ is the restriction of C to E in the fixed coordinate convention.

The second row follows from

$$(2, 2m-1, 2)A_E = (2h+4m+2, (2m-1)^2+4m, 4m),$$

and $(2m-1)^2+4m = 4m^2+1$. The third row is

$$(0, 0, h)A_E = (2h, 2mh, h).$$

These calculations fix every entry relative to (7.6); changing the meaning of b from a common coordinate to a sum of middle coordinates would produce a different matrix.

Proposition 7.1 (Invariant splitting). *The sum (7.5) is invariant under A , B , and C . Their restrictions to U are given by (7.3); their restrictions to E are (7.7), (7.8), and (7.9).*

Proof. The calculation on U , the characteristic-zero intersection argument, and the dimension count prove the direct sum. Middle permutation symmetry proves invariance of E , and the displayed row calculations give the three restriction matrices. $\square$

7.3 Trace, principal minors, and determinant

The trace is

$$\text{tr } Q_{m,h} = 2h + 4m^2 + 4m + 5. \quad (7.10)$$

The principal 2×2 minors for coordinate pairs 12, 13, and 23 are

$$\Delta_{12} = h(1 - 4m(m+1)) + 4, \quad (7.11)$$

$$\Delta_{13} = h(h - 4m), \quad (7.12)$$

$$\Delta_{23} = h(1 - 4m^2). \quad (7.13)$$

For instance,

$$\Delta_{13} = h(h + 4m + 4) - 4h(2m + 1) = h(h - 4m);$$

the other two follow from the corresponding two-term determinants. Their sum is

$$\Delta_{12} + \Delta_{13} + \Delta_{23} = h^2 - 8hm(m+1) + 2h + 4. \quad (7.14)$$

For transcription control, the other two expansions are

$$\begin{aligned}\Delta_{12} &= (h + 4m + 4)(4m^2 + 1) - 2m(2m + 1)(2h + 4m + 2) \\ &= h(1 - 4m(m + 1)) + 4, \\ \Delta_{23} &= h(4m^2 + 1) - 4m(2mh) = h(1 - 4m^2).\end{aligned}$$

Adding these to Δ_{13} produces (7.14) without appealing to a determinant expansion of the full $r \times r$ matrix. The determinant is clearest before multiplication:

$$\det A_E = -h(2m + 1), \quad \det B_E = -h(2m + 1). \quad (7.15)$$

Indeed, expansion along the first row of A_E and the last row of B_E , respectively, gives

$$\det A_E = h((2m - 1) - 4m), \quad \det B_E = h((2m - 1) - 4m).$$

Both determinants are therefore nonzero in K , although only their product is needed for the characteristic polynomial of Q . Therefore

$$\det Q_{m,h} = h^2(2m + 1)^2. \quad (7.16)$$

Table 3: Three-dimensional invariant ledger for $Q_{m,h}$.

Invariant	exact value	role in $\det(tI - Q)$
trace	$2h + 4m^2 + 4m + 5$	negative t^2 -coefficient
principal-minor sum	$h^2 - 8hm(m + 1) + 2h + 4$	t -coefficient
determinant	$h^2(2m + 1)^2$	negative constant term

These invariants give

$$\begin{aligned}\det(tI_3 - Q_{m,h}) = P_{m,h}(t) &= t^3 - (2h + 4m^2 + 4m + 5)t^2 \\ &\quad + (h^2 - 8hm(m + 1) + 2h + 4)t \\ &\quad - h^2(2m + 1)^2.\end{aligned} \quad (7.17)$$

On U , the characteristic polynomial is $(t - 1)^{r-3}$. The direct sum therefore yields

$$\chi_C(t) = (t - 1)^{r-3} P_{m,h}(t). \quad (7.18)$$

7.4 The value at one and exact unit multiplicity

Substitution into (7.17) gives

$$\begin{aligned}P_{m,h}(1) &= 1 - (2h + 4m^2 + 4m + 5) \\ &\quad + (h^2 - 8hm(m + 1) + 2h + 4) - h^2(2m + 1)^2 \\ &= -4m(m + 1)h^2 - 8m(m + 1)h - 4m(m + 1) \\ &= -4m(m + 1)(h + 1)^2.\end{aligned} \quad (7.19)$$

This is nonzero in characteristic zero. Hence $C|_E$ has no unit eigenvalue. All generalized unit eigenvectors lie in U , where C is exactly the identity.

Theorem 7.2 (Exact unit multiplicity). *The eigenvalue 1 of C has algebraic and geometric multiplicity exactly $r - 3$. There is no quotient unit eigenvalue and no hidden unit Jordan block.*

Proof. The direct factorization gives at least the $r - 3$ unit roots contributed by U . Equation (7.19) excludes another root on E . Because $C|_U = I_U$, the unit factor on U is semisimple. Both multiplicities are therefore $\dim U = r - 3$. $\square$

7.5 The cubic annihilator

The seed $\mathbf{1}$ belongs to E , represented by $(1, 1, 1)$ in (7.6). Define

$$\begin{aligned} T_0 &= 2h + 4m^2 + 4m + 5, \\ S_0 &= h^2 - 8hm(m + 1) + 2h + 4, \\ D_0 &= h^2(2m + 1)^2. \end{aligned} \tag{7.20}$$

Cayley–Hamilton on E gives

$$Q_{m,h}^3 - T_0 Q_{m,h}^2 + S_0 Q_{m,h} - D_0 I_3 = 0.$$

Applying this identity to the orbit of $\mathbf{1}$ and then applying $e_r^\top$ proves:

In the convention (7.6), the seed is the column $(1, 1, 1)^\top$, the action of C is $Q_{m,h}$, and restriction of $e_r^\top$ is projection onto the third coordinate. Consequently,

$$d_n = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} Q_{m,h}^n \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Multiplying the matrix identity by the seed column on the right and the third-coordinate row on the left gives the scalar relation with exactly the same signs as $P_{m,h}$.

The direct-sum viewpoint matters here. Cayley–Hamilton for the full matrix C would give an order- r relation containing the unit factor. Since the seed has equal middle coordinates, its U -component is zero, and its entire orbit stays in E . The cubic identity therefore annihilates the degree orbit directly, with no residual term from the $(r - 3)$ -dimensional unit sector.

Theorem 7.3 (Cubic degree recurrence). *For*

$$d_n = e_r^\top C^n \mathbf{1} = \deg(F_{r,g}^n),$$

one has

$$d_{n+3} = T_0 d_{n+2} - S_0 d_{n+1} + D_0 d_n \quad (n \geq 0), \tag{7.21}$$

with

$$d_0 = 1, \quad d_1 = h(2m + 3). \tag{7.22}$$

Proof. The recurrence is the scalar image of Cayley–Hamilton. The initial value $d_0 = 1$ is immediate. The last row of C has $m + 1$ entries equal to $2h$ and one entry equal to h , so

$$d_1 = 2h(m + 1) + h = h(2m + 3).$$

□

The cubic is an annihilator. It is not asserted to be minimal or irreducible for every parameter, and $\rho(C)$ need not always have algebraic degree three. The Perron class nevertheless belongs to E : C commutes with middle-coordinate permutations, and uniqueness of its positive Perron ray forces that ray to be fixed by every such permutation. Thus $\rho(C)$ is a root of $P_{m,h}$ without an irreducibility claim.

8 Sharp boundary, low-mode consistency, fixed-support coefficients, limitations, and conclusion

The theorem has three delicate edges: the first parameter excluded from the strict seed cone, the formal three-mode substitution, and changes of coefficients that preserve every support. Each edge is an exact calculation, not an invitation to broaden the theorem.

8.1 The $g = 2r$ seed/selected-face boundary

At the ordinary-degree seed, the first pure support has score $h = g - 1$. The competing mixed row has score

$$1 + 2(r - 1) = 2r - 1.$$

Their difference is exactly

$$(g - 1) - (1 + 2(r - 1)) = g - 2r. \quad (8.1)$$

When $g = 2r$, the two scores tie. At the same parameter,

$$\sigma(\mathbf{1}) = r - 1 = \frac{g - 2}{2}, \quad (8.2)$$

so the seed lies on the boundary of the open height face in (3.6). These are two descriptions of the same obstruction to starting the present strict selection proof.

At the first allowed value $g = 2r + 1$, the score gap (8.1) is one. Since then $h = 2m + 4$, the height slack is

$$\frac{h - 1}{2} - (m + 1) = \frac{2m + 3}{2} - (m + 1) = \frac{1}{2}.$$

All later factored margins contain $h - 2m - 3 = 1$, so the proof is strict from its first admissible parameter onward.

Proposition 8.1 (Boundary statement). *The inequality $g \geq 2r + 1$ is sharp for the ordinary seed and the chosen strict selected face: at $g = 2r$, the first selector ties and the seed lies on the open cone-height boundary.*

Proof. Equations (8.1) and (8.2) give the two claims. The gap-one and half-unit-slack calculations establish strictness at $g = 2r + 1$. $\square$

Proposition 8.1 does not determine the actual degree dynamics at $g = 2r$ or below. A different selector region, a tie analysis, or a different observable could behave differently. Accordingly, the proposition is not a global failure theorem and does not claim that the parameter range is optimal for every possible degree description.

8.2 Formal low-mode consistency

The headline theorem begins at $r \geq 4$. A useful transcription test is to substitute formally $r = 3$, hence $m = 1$, only after the arbitrary-mode proof is complete. With $h = g - 1$, (7.9) becomes

$$Q_{1,g-1} = \begin{pmatrix} g+7 & 6 & 6 \\ 2g+4 & 5 & 4 \\ 2(g-1) & 2(g-1) & g-1 \end{pmatrix}. \quad (8.3)$$

The trace coefficient specializes to $2g + 11$. The principal-minor sum is

$$(g - 1)^2 - 16(g - 1) + 2(g - 1) + 4 = g^2 - 16g + 19,$$

and the determinant is $9(g - 1)^2$. Therefore

$$P_{1,g-1}(t) = t^3 - (2g + 11)t^2 + (g^2 - 16g + 19)t - 9(g - 1)^2. \quad (8.4)$$

Equations (8.3) and (8.4) constitute formal low-mode consistency only. They lie outside the stated range $r \geq 4$, prove no additional $r = 3$ theorem, and do not alter the headline result.

8.3 Four nonzero coefficients on the fixed supports

The support proof is insensitive to nonzero scalar changes but not to support changes. Let

$$\begin{aligned} V_{\alpha,\beta}(q) &= \alpha \prod_{i=1}^r q_i^2 + \beta q_1^g, \\ W_{\gamma,\delta}(p) &= \gamma \prod_{i=1}^r p_i^2 + \delta p_r^g, \end{aligned} \quad \alpha, \beta, \gamma, \delta \in K^\times. \quad (8.5)$$

Define the two gradient shears and their composition in the same order as before.

The derivative scalars attached to the four displayed support types are

$$2\alpha, \quad g\beta, \quad 2\gamma, \quad g\delta.$$

They are nonzero because the four coefficients are nonzero and K has characteristic zero. Thus every exponent row in Table 1 remains present. Selector margins compare degrees of distinct supports and are unchanged; the cone, its wall inequalities, A , B , and C are therefore unchanged as well.

It remains to justify arbitrary signs or phases of the coefficients. At each coordinate update, the strict selector and carry inequalities identify one source of uniquely highest degree. Its highest homogeneous form is one of the nonzero scalars above times a product of powers of preceding nonzero highest homogeneous forms. The polynomial ring over K is a domain, so that product is nonzero. Because every other source has smaller degree, none can cancel it. This is the same induction as in Lemma 5.2, and it never uses coefficient positivity.

Corollary 8.2 (Fixed-support coefficient extension). *For the potentials in (8.5), with $\alpha, \beta, \gamma, \delta \in K^\times$, the phase recurrence, exact degree identity, first dynamical degree, invariant splitting, quotient matrix, characteristic factorization, exact unit multiplicity, and cubic annihilator are the same as in Theorem 1.1.*

Proof. The gradient shears retain subtraction inverses and symmetric Hessian blocks. The four derivative scalars are nonzero, so the literal supports and every strict comparison are unchanged. The domain leading-form argument proves noncancellation at every phase. All subsequent conclusions depend only on the resulting matrices and the visible degree identity. $\square$

If one coefficient vanishes, its support can disappear. If a monomial is added, a new competitive face can appear. An altered support, a different shear word, or a positive-characteristic specialization therefore falls outside Corollary 8.2.

8.4 Limitations

The selector cone is explicit, sufficient, and invariant, but no maximality, necessity, uniqueness, or classification statement is proved. The theorem fixes both supports and the word $T \circ S$; it does not cover arbitrary potentials, added monomials, arbitrary Hamiltonian maps, or other shear words. Characteristic zero is used to keep derivative coefficients and the integer m nonzero. Strict last-coordinate visibility begins at $n = 1$, while $n = 0$ remains a tie. The threshold calculation is sharp only at the ordinary seed and the selected strict face. Finally, the cubic annihilates the degree sequence but need not be minimal or irreducible, so no uniform algebraic-degree-three conclusion follows for the Perron root.

8.5 Conclusion

For every characteristic-zero field, $r \geq 4$, and $g \geq 2r + 1$, the displayed endpoint-spiked Hamiltonian product shears have the exact degree and Perron identities

$$\deg(F_{r,g}^n) = e_r^\top C^n \mathbf{1}, \quad \lambda_1(F_{r,g}) = \rho(C).$$

The r -dimensional degree system splits as an $(r - 3)$ -dimensional semisimple unit space and a three-dimensional equal-middle restriction. Its characteristic polynomial is

$$(t - 1)^{r-3} P_{m,h}(t),$$

and $P_{m,h}$ supplies the exact cubic annihilating recurrence for the degree sequence. Increasing the number of modes enlarges the unit sector without enlarging the nontrivial spectral dimension of this fixed family.

References

- Jérémy Blanc and Immanuel van Santen. Dynamical degrees of affine-triangular automorphisms of affine spaces, 2021. URL <https://arxiv.org/abs/1912.01324>.
- Nguyen-Bac Dang and Charles Favre. Spectral interpretations of dynamical degrees and applications. *Annals of Mathematics*, 194, 2021. doi: 10.4007/annals.2021.194.1.5. URL <https://doi.org/10.4007/annals.2021.194.1.5>.
- Julie Déserti. Degree growth of polynomial automorphisms and birational maps: some examples, 2016. URL <https://arxiv.org/abs/1602.04642>.
- Keisuke Fujioka, Ryota Kogawa, Jizhou Li, and Akira Shudo. Coupled hénon map, part i: Topological horseshoes and uniform hyperbolicity, 2023. URL <https://arxiv.org/abs/2303.05769>.
- G. Rangarajan. Polynomial map symplectic algorithm, 2002. URL <https://arxiv.org/abs/physics/0212098>.
- Enbo Shao and Xiaosong Sun. Dynamical degrees of affine-triangular automorphisms in dimension four, 2025. URL <https://arxiv.org/abs/2509.14584>.